

IoT-Based Real-Time Tide Monitoring Tool: Design and Case Study at Kenjeran Beach, Surabaya

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The dynamic nature of tides poses significant challenges for coastal management and navigation, particularly in regions with high maritime activity. This study focuses on the design and implementation of an IoT-based real-time tide monitoring system, used in a case study at Kenjeran Beach, Surabaya. The methodology involved three main phases: system design, hardware implementation, and software integration. The hardware setup included ultrasonic water level sensors, Arduino-based microcontrollers, and NRF24L01 for data transmission. On the software side, a cloud platform was utilized for data storage, visualization, and alert mechanisms. The system was designed to be energy-efficient and capable of operating in remote coastal areas with minimal maintenance. The system utilized a NodeMCU ESP8266 microcontroller, an NRF24L01 wireless data communication module, and the Blynk application for real-time monitoring via smartphones. The system consisted of a sensor module that measured tidal data using NRF24L01 and a server module that transmitted the data via NodeMCU ESP8266. The sensor module used an SRF05 sensor to measure water surface height and converted it into water data. Field tests were conducted at Kenjeran Beach to evaluate the accuracy, reliability, and resilience of the system in real-world conditions. The results demonstrated that the system can accurately measure tidal variations with a margin of error below ± 1.3175 cm and provide real-time data updates every fifteen minutes. Sensor data accuracy was tested and found to be 99.554%, with maximum distance measurement of 330 cm and an error rate of 1.43%. The Blynk application was tested for seven days and showed that the Kenjeran Beach Tourism Area experiences two high tides and two low tides in 24 hours. This research highlights the potential of IoT technology for advancing real-time environmental monitoring and offers valuable insights for stakeholders in coastal zone management, fisheries, and maritime navigation.

KEYWORDS